RESEARCH COMPILATION 2026

The Learning Brain

Dopamine as TD error, predictive processing, hippocampal replay, grid cells, global workspace: what the brain teaches machine builders.

Intelligence Research Atlas - Volume 04 of 6 -

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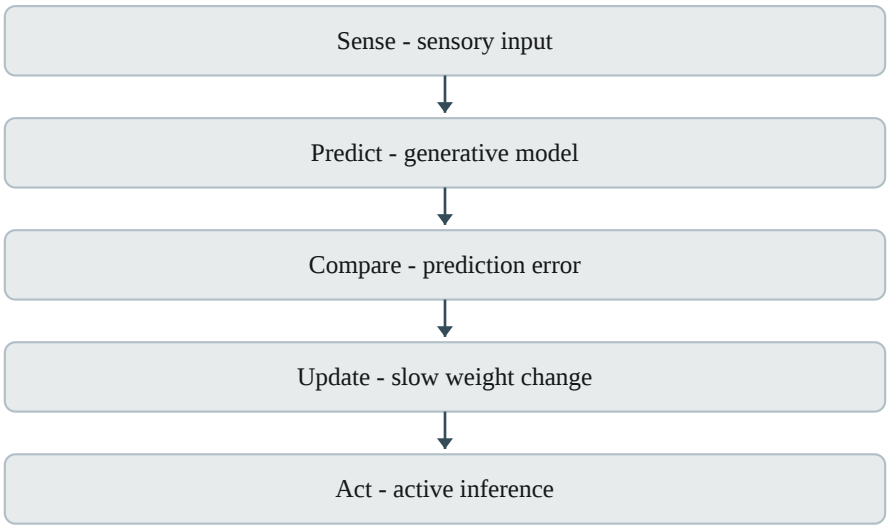
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The Neuroscience of Intelligence

26 papers bridging cortex and compute, distilled into 14 findings - the parts catalog for intelligence designs that backprop alone cannot supply.

The Learning Brain - the through-line



Part I - Distilled Findings

F1 - Neurons that fire together wire together: local learning builds global structure

Implication for building intelligence: Biology computes statistics with local rules - no global loss function, no error backprop through a deep stack. Local-rule learning (Hebbian, anti-Hebbian, predictive-coding local errors) is the enabling technology for neuromorphic hardware and lifelong learning systems that cannot afford end-to-end backprop.

F2 - Dopamine IS the TD error: the brain runs reinforcement learning

Implication for building intelligence: Validation that RL is not merely "inspired by" brains - it IS the brain's evaluation subsystem. When engineering reward (RLHF, RLVR, domain 03), expect biological pathologies: reward hacking ≈ addiction, over-training on proxy reward ≈ dopamine dysregulation.

F3 - The brain is a prediction machine: perception as controlled hallucination

Implication for building intelligence: Strongest convergence result in neuro-AI: next-token prediction loss and cortical predictive coding are the same design pattern at different substrates. Predictive-coding local errors are also a live backprop alternative (Salvatori et al. 2025 survey).

F4 - Sparse, distributed, high-dimensional coding - twice over

Implication for building intelligence: Convergent evolution across substrates suggests sparse distributed coding is near-optimal for capacity/generalization tradeoffs. If designing representations or memory formats, default to sparse high-dimensional codes; use dictionary learning (SAEs) to audit them.

F5 - Hippocampus: relational maps, grid codes, replay - the memory subsystem design

Implication for building intelligence: The blueprint for agent memory: episodic store + spatial/relational indexing + prioritized replay for consolidation and planning. DQN's replay buffer was the first loan; agentic systems are still under-collecting on this design (domain 05).

F6 - Sleep is training infrastructure: consolidation by renormalization

Implication for building intelligence: Continual-learning systems should schedule consolidation phases (replay + renormalization + pruning) between acquisition phases - systems that only ever accumulate weights degrade (catastrophic forgetting). "Sleep" for your agents is a real design pattern.

F7 - Global workspace: bottlenecked broadcast as an architecture

Implication for building intelligence: A ready-made pattern for multi-agent LLM systems: specialists + arbiter + shared context. Also a candidate theory of conscious access - useful both as architecture and as a lens on system "attention".

F8 - Brains operate at criticality: the edge of chaos

Implication for building intelligence: Initialization/dynamics lore in deep learning (edge-of-chaos RNNs, signal propagation) is the same mathematics. If your recurrent/feedback system is either too stable (dead) or too chaotic (untrainable), tune toward criticality deliberately.

F9 - The brain's wiring is small-world and modular - topology is compute

Implication for building intelligence: When distributing an AI system across modules or machines, small-world design (dense local, sparse global) is the default-robust topology; also a maintenance principle: keep interfaces narrow (like white-matter bundles).

F10 - Backprop's biological gap has working alternatives

Implication for building intelligence: For energy-efficient / lifelong / analog substrates (neuromorphic, optical, biological), local learning rules are the enabling constraint - this research line is the toolbox.

F11 - Neuromodulation is global state control - hyperparameters as signals

Implication for building intelligence: In ML terms: learned global conditioning - adaptive optimizers, learned exploration schedules, context-dependent gain (as in hypernetworks and meta-learning). Multi-agent designs should include a dedicated, low-bandwidth state-control channel rather than burying mode-switches in weights.

F12 - Representational geometry is the shared language of brains and networks

Implication for building intelligence: RSA-style comparison is a practical tool: benchmark your model's internal geometry against known-good systems (brain or model), detect pathological representations early, and choose regularizers that shape geometry (not just loss).

F13 - Marr's three levels keep AI research honest

Implication for building intelligence: Use Marr as a review discipline: when a paper claims an ability, ask which level the evidence actually supports. When designing systems, decide the computational problem FIRST - most failed architectures are level-mismatches (right problem, wrong bias).

F14 - The street runs both ways: neuro->AI and AI->neuro

Implication for building intelligence: Treat this corpus as a two-way toolkit: before building a component, check the neuro literature for a proven mechanism; before theorizing about brains, build the model and test it. The fastest path to novel intelligence runs through this loop deliberately, not accidentally.

Part II - The Canon

26 landmark papers in chronological order. Full entries with links and keywords live in the corpus folder 04_neuroscience-of-intelligence/papers.md.

Year	Paper / Authors	Core idea	Why it matters
1952	**A Quantitative Description of Membrane Current and Its Application to Conduction and Excitation in Nerve** Alan Hodgkin, Andrew Huxley	Coupled differential equations reproducing the action potential from ion-channel dynamics.	Proof that computation can be *derived* from biophysics - the template for mechanistic explanation that AI's interpretability now imitates.
1962	**Receptive Fields, Binocular Interaction and Functional Architecture in the Cat's Visual Cortex** David Hubel, Torsten Wiesel	Cortical neurons are arranged in hierarchical feature detectors - simple cells feed complex cells with invariance.	The direct biological inspiration for CNNs (LeCun 1989) - hierarchy + feature + invariance was observed in cortex 27 years before AlexNet.

Year	Paper / Authors	Core idea	Why it matters
1978	**The Hippocampus as a Cognitive Map** John O'Keefe, Lynn Nadel	Place cells encode location; hippocampus implements a map-like relational memory, not a warehouse.	Memory as structured, relational, spatial - the design reference for vector stores, agent spatial reasoning, and episodic memory architectures.
1982	**Vision: A Computational Investigation into the Human Representation and Processing of Visual Information** David Marr	Any intelligent system must be understood at three levels - computation (what/why), algorithm (how), implementation (substrate) - and levels are separable.	The discipline that keeps AI honest: transformer = one algorithmic answer; LLM scaling debates are arguments at Marr's computational level.
1994	**Reactivation of Hippocampal Ensemble Memories During Sleep** Matthew Wilson, Bruce McNaughton	Sequences of place-cell activity from waking experience are replayed during sleep.	Offline consolidation as a computation - the biological justification for experience replay in RL (DQN) and for offline training phases generally.
1996	**Emergence of Simple-Cell Receptive Field Properties by Learning a Sparse Code for Natural Images** Bruno Olshausen, David Field	Optimizing for SPARSE codes of natural images reproduces V1 receptive fields - no teacher, no labels.	The brain's sensory code is shaped by efficient coding of environmental statistics; the direct ancestor of sparse autoencoder features in LLM interpretability (domain 03, F8).
1997	**A Neural Substrate of Prediction and Reward (dopamine neurons)** Wolfram Schultz, Peter Dayan, Read Montague	Dopamine neurons do not signal reward - they signal REWARD PREDICTION ERROR, exactly the TD error of Q-learning.	The single most celebrated neuro<->AI bridge: a learning algorithm (TD, domain 02) was discovered in neurons before it was standard in machines (or rather, matched after) - reward prediction errors drive both.
1999	**Predictive Coding in the Visual Cortex: A Functional Interpretation of Some Extra-Classical Receptive-Field Effects** Rajesh Rao, Dana Ballard	Cortex transmits prediction ERRORS up the hierarchy (feedback carries predictions); explains end-stopping and contextual effects.	Bandwidth-efficient, learned, hierarchical inference - the brain runs a self-supervised predictive model; the blueprint for predictive-processing architectures and a candidate biologically plausible learning rule.
2003	**A Neural Global Workspace Theory of Conscious (and Unconscious) Processing** Stanislas Dehaene, Jean-Pierre Changeux (building on Baars' Global Workspace 1988)	Conscious access = broadcasting of a winner-take-all selection through a global workspace - a shared, capacity-limited bottleneck routed to many processors.	A concrete architecture pattern (bottleneck + broadcast) that inspired MoE-style routing, LLM context as workspace, and modular agent designs (Goyal & Bengio 2020).

Year	Paper / Authors	Core idea	Why it matters
2004	**An Information-Theoretic Measure of Consciousness (Integrated Information Theory)** Giulio Tononi	Consciousness corresponds to integrated information (Φ) - the causal irreducibility of a system's parts.	Influential formal program linking substrate structure to experience; CONTested (adversarial testing 2023 favored recurrent processing over IIT) - include as theory-design lesson: formal elegance ≠ empirical validity.
2004	**Neuronal Circuits of the Neocortex: The Canonical Microcircuit** Rodney Douglas, Kevan Martin	Cortex is a repeated canonical circuit (6 layers, characteristic recurrence and inhibition) whose connectivity is stereotyped across areas.	Nature's answer to architecture reuse: one repeated block, specialized by input/output, prefigures layer-stacking universality (transformer blocks).

Year	Paper / Authors	Core idea	Why it matters
2005	**Microstructure of a Spatial Map in the Entorhinal Cortex (grid cells)** Torkel Hafting, Marianne Fyhn, Sturla Molden, May-Britt Moser, Edvard Moser	Entorhinal grid cells fire on hexagonal tessellations - a metric spatial code.	Nobel-recognized discovery of an elegant internal coordinate system; grid-cell-like codes emerge in RL agents (Banino 2018) - convergent evolution of representation between brains and networks.
2005	**Uncertainty, Neuromodulation, and Attention (ACh & NE)** Angela Yu, Peter Dayan	Acetylcholine signals expected uncertainty; norepinephrine signals unexpected uncertainty - neuromodulators encode uncertainty types, gating learning and attention.	Global state control as architecture: neuromodulation is a broadcast hyperparameter system - the biological template for meta-learners, adaptive optimizers, and context conditioning in ML.
2006	**Small-World Brain Networks** Danielle Bassett, Ed Bullmore	Human brain graphs combine high local clustering with short global path lengths - optimal for segregated AND integrated processing.	Network-architecture lesson for AI: mixture-of-experts routing, agent topologies, and MoE load-balancing are engineering rediscoveries of small-world design.
2006	**Neuronal Avalanches in Neocortical Circuits** John Beggs, Dietmar Plenz	Cortical activity propagates in power-law-distributed avalanches - signatures of operation near a critical point.	Evidence the brain self-tunes to criticality (maximal dynamic range, information transmission) - matching self-organized criticality in physics (domain 01, F11) and edge-of-chaos results in RNN/attention dynamics.
2006	**Sleep and Synaptic Homeostasis** Giulio Tononi, Chiara Cirelli	Sleep globally downscales synaptic strengths - renormalizing after a day of potentiation, preserving relative weights.	A biological analogue of normalization and weight decay - "forget the noise, keep the signal"; a design hint for continual-learning systems that consolidate by re-normalizing.
2008	**Representational Similarity Analysis (RSA): Connecting Computational Models with Brains** Nikolaus Kriegeskorte, Marieke Mur, Peter Bandettini	Compare representations via geometry of pairwise distances - model-agnostic comparison between brains, behavior, and machines.	The measurement tool of representational convergence: CNN layers <-> visual cortex correspondences (later Yamins et al. 2014) were established this way; reusable for comparing your model's internal geometry to anything.
2016	**Random Feedback Weights Support Learning in Deep Neural Networks (feedback alignment)** Timothy Lillicrap, Daniel Cownden, Douglas Tweed, Colin Akerman	Backward weights need not be symmetric - networks adapt their forward weights to whatever feedback channel exists.	Kills the "backprop is biologically impossible" objection in its simplest form; enables training on hardware where exact reverse passes are impractical.

Year	Paper / Authors	Core idea	Why it matters
2016	**Towards an Integration of Deep Learning and Neuroscience** Adam Marblestone, Greg Wayne, Konrad Kording	Systematic map of what neuroscience could contribute to deep learning: memory, attention, continual learning - with the transfer direction specified.	The reference agenda for neuro->AI borrowing; use as a checklist against your architecture design.
2017	**Dendritic Error Backpropagation in Deep Cortical Microcircuits** João Sacramento, Rui Ponte Costa, Yoshua Bengio, Walter Senn	Layer-specific dendritic compartments can carry error signals locally, approximating backprop in a biologically plausible cortical circuit.	Existence proof that error-driven deep learning might be implementable in wetware - and a template for local-learning-rule hardware (neuromorphic, optical).

Year	Paper / Authors	Core idea	Why it matters
2017	**Neuroscience-Inspired Artificial Intelligence** Demis Hassabis, Dharshan Kumaran, Courtney Summerfield, Matthew Botvinick	Historical and forward synthesis: reliving the neuro->AI pipeline that produced CNNs/RL/attention, and what remains untapped (meta-learning, memory consolidation).	The manifesto of the field from DeepMind's cofounder - argues AGI likely requires re-engagement with brain mechanisms.
2018	**Prioritized Memory Access Reveals Planning Networks in the Brain** Marcelo Mattar, Nathaniel Daw	What to replay is itself a computation: expected reward gain × shortcut-to-decision determine replay priority.	Explains WHY experience replay works and when: prioritized replay is information-theoretically optimal curriculum - directly transferable to RL buffer design and agent memory management.
2018	**Prefrontal Cortex as a Meta-Reinforcement Learning System** Jane Wang, Zeb Kurth-Nelson, Dharshan Kumaran, ... Matthew Botvinick, Demis Hassabis	The brain's prefrontal system implements meta-RL: slow synaptic learning (outside) shapes a fast in-context learner (inside) - and the same dynamics emerge in recurrent ANNs trained on task distributions.	Unifies biological and artificial in-context learning: prefrontal meta-RL ≈ transformer ICL (domain 03, F7); tells you when learned recurrent state beats weight updates.
2019	**High-Dimensional Geometry of Population Responses** Stringer, C. et al. (Carsen Stringer, Marius Pachitariu, Nicholas Steinmetz, ... Kenneth Harris)	Cortical populations track many stimulus variables (near-power-law eigenspectrum) while remaining decodable - high-dimensional but smooth, no collapse.	Empirical geometry that neural networks should match; connects to superposition (domain 03, F8) and neural collapse (domain 01) - three views of representation geometry across substrates.
2020	**Inductive Biases for Higher-Level Cognition** Anirudh Goyal, Yoshua Bengio	Transplant Global-Workspace-style inductive biases (bottleneck, modules, broadcast, attention control) into deep nets for combinatorial generalization.	The most concrete GWT-to-architecture pipeline - directly relevant to modular/multi-agent LLM system design (domain 05).
2025	**A Survey on Neuro-Mimetic Deep Learning via Predictive Coding** Tommaso Salvatori et al.	Comprehensive survey of predictive coding as a learning algorithm - approximating backprop with local errors, with efficiency gains on edge hardware.	The 2025 state of a serious backprop alternative: local, online, event-friendly - candidate learning rule for neuromorphic and continually-learning systems (domain 05).

Full sources with links live in the corpus: papers.md / papers.json per domain.

Where to go next

The living version of this canon lives in the corpus folder for this domain: papers.md carries every entry with full links and keywords, findings.md holds the complete distilled findings, and the diagrams folder contains Mermaid sources that render in Obsidian and GitHub. To see how this domain connects to the rest of the atlas, read the cross-domain map in 00_master-atlas; to extend the corpus, follow the conventions in AGENTS.md and re-run the index and PDF build scripts in tools/.